Cypher Stack

Lecture 2 - Divisibility and Proofs

What we need to understand it

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Outline



1. Modular Arithmetic

2. Divisibility

3. Proving Things

Section 1

Modular Arithmetic

Modular Properties



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Modular Properties



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We only need to discuss one more property regarding modular arithmetic in order to have all the tools necessary in order to develop our first key exchange protocol.

Modular Arithmetic

Exponent Laws



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Modular Arithmetic

Exponent Laws



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Definition 1 (General Exponent Law)

For all $a, x, y \in \mathbb{N}$, the following are true:

$$a^{x+y} = a^x \cdot a^y$$

$$(a^x)^y = (a^y)^x = a^{xy}$$

Modular Arithmetic

Exponent Laws



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Modular Arithmetic

Exponent Laws



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Now let's shift our view to a concept intricately connected to modular arithmetic: divisibility.

Section 2

Divisibility

Divisibility



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Divisibility



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From here on out, we will use special notation to represent this, which is as follows:

Divisibility

Definition



Definition 2

Divisibility Let $a, b \in \mathbb{N}$. We say that a divides b, given that there exists some k such that:

$$b = a \cdot k$$

We denote this by the notation:

$$a \mid b$$

Divisibility

The Division Algorithm



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Divisibility

The Division Algorithm



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Definition 3 (The Division Algorithm)

Let $a, b \in \mathbb{Z}$, where $b > 0$. There exists unique integers, q and r, such that:

$$a = b \cdot q + r \ , \ 0 \leq r < b$$

We call q the *quotient* and r the *remainder*.

Divisibility

The Bridge to Modular Arithmetic



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The Bridge to Modular Arithmetic



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Divisibility

The Bridge to Modular Arithmetic



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We can remember that modular arithmetic is just '*traveling around the clock*', where each time we pass the modulus, we go to zero.

This isn't all too different from dividing a number by another number. In fact, it is more or less the same, just presented in a different way.

Divisibility

The Bridge to Modular Arithmetic



Let's look at an example:

Divisibility

The Bridge to Modular Arithmetic



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Divisibility

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Well, 25 is that number.

Divisibility

The Bridge to Modular Arithmetic



So, let's count:

25	26	27
0	1	2

Divisibility

The Bridge to Modular Arithmetic



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The Bridge to Modular Arithmetic



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So, we can see that:

$$27 \equiv 2 \mod 5$$

Divisibility

The Reveal



But wait a minute. Let's do a little bit of algebraic rearrangement:

$$27 - 25 = 2$$

Divisibility

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But wait a minute. Let's do a little bit of algebraic rearrangement:

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$$27 = 5 \cdot 5 + 2$$

This looks like the form of the division algorithm! In this case, the $a = 27$ (our original number) and the $b = 5$ (our modulus).

Divisibility

The Second Reveal



So, it appears to be the case that if we apply the division algorithm, using our modulus as b, then the *remainder* happens to be equal to it mod b!

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So, it appears to be the case that if we apply the division algorithm, using our modulus as b, then the *remainder* happens to be equal to it mod b!

As it turns out, this is exactly the correct conclusion: modular arithmetic gives us the *remainder* on division by the modulus!

Divisibility

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This is a useful connection to make, since first and foremost, it allows us to reduce numbers much more efficiently.

More importantly, it tells us that the properties of modular arithmetic are intricately intertwined with divisibility!

Section 3

Proving Things

Proof



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The heart of mathematics lies in mathematical proof: where you claim a statement is true and through a series of valid, logical steps, provide a rational argument of your claim.

The proof of a statement is an exact mathematical explanation of why the statement holds, so that anyone with the patience can believe the claim without relying on believing the person who wrote it.

Proof



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The art of writing mathematical proofs is a convoluted one, but we will start with some simple example to see how it works.

Proof

Example 1



Let's start with something simple and related to divisibility!

Proof

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Let's start with something simple and related to divisibility! Let's prove the following statement:

Let $a, b \in \mathbb{N}$. If $2 \mid a$ and $2 \mid b$, then $2 \mid a + b$.

Proof

Example 2



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It seems to me that, in the proof we just did, we can add a new hypothesis and get a valid result. Let's try to prove this new, related result:

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It seems to me that, in the proof we just did, we can add a new hypothesis and get a valid result. Let's try to prove this new, related result:

Let $a, b, x, y \in \mathbb{N}$. If $2 \mid a$ and $2 \mid b$, then $2 \mid ax + by$.

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Example 3



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Can you see what it is?

If you asked the question: why did we choose the number 2 instead of some other natural number, that is exactly the right question to ask!

Proof

Example 3



So let's prove the most general form of this statement:

Let $a, b, d, x, y \in \mathbb{N}$. If $d \mid a$ and $d \mid b$, then $d \mid ax + by$.

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This is the most common type of proof. However, it is not the only type of proof.

The Second Type

Contrapositive



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Alternatively, you can assume the negation of Q and reason until you get to the negation of P. These are logically equivalent and in this form, it is called **proof by contrapositive**.

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Sometimes it is easier to approach a problem by contrapositive than it is directly. Let's look at an example.

Contrapositive

Example 1



Let's try to prove the following:

If n^2 is even, then n is even.

My Personal Favorite

Proof By Contradiction



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The idea here is simple: if you want to prove a statement S, you could assume that the negation of S is true, and then cause something impossible to happen.

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Proof By Contradiction



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The idea here is simple: if you want to prove a statement S, you could assume that the negation of S is true, and then cause something impossible to happen.

If the negation of something is false (by virtue of making something impossible happen), then the original statement must be true.

Contradiction

Why use it?



Proof by contradiction is a powerful tool because it does not rely on your ability to find the line of reasoning that leads from the hypotheses to the conclusion, rather, it relies on your ability to determine when something mathematically 'bad' is happening.

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Proof by contradiction is a powerful tool because it does not rely on your ability to find the line of reasoning that leads from the hypotheses to the conclusion, rather, it relies on your ability to determine when something mathematically 'bad' is happening.

Often times, finding the impossible thing can be easier than the alternatives. Let's look at some examples.

Contradiction

Example 1



Prove that there are infinitely many prime numbers.